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Review – Prostate Cancer – Editor's choice

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Magnetic Resonance Imaging–based Biopsy Strategies in Prostate Cancer Screening: A Systematic Review

Ivo G. Schoots [a,b,*], *Hashim U. Ahmed* [c,d], *Peter Albers* [e,f], *Patrick Asbach* [g], *Roderick C.N. van den Bergh* [h], *Rebecka A. Godtman* [i,j], *Pim J. van Leeuwen* [b,k], *Tobias Nordström* [l,m], *Shonit Punwani* [n], *Jonas Wallström* [o,p], *Anwar R. Padhani* [q]

[a] *Department of Radiology and Nuclear Medicine, Erasmus University Medical Center, Rotterdam, The Netherlands;* [b] *Center of Early Diagnosis, Netherlands Cancer Institute, Amsterdam, The Netherlands;* [c] *Imperial Prostate, Division of Surgery, Department of Surgery and Cancer, Imperial College London, London, UK;* [d] *Imperial Urology, Division of Cancer, Cardiovascular Medicine and Surgery, Imperial College Healthcare NHS Trust, London, UK;* [e] *Department of Urology, University Hospital, Medical Faculty, Heinrich-Heine University Duesseldorf, Duesseldorf, Germany;* [f] *Division of Personalized Early Detection of Prostate Cancer, German Cancer Research Center (DKFZ), Heidelberg, Germany;* [g] *Department of Radiology, Charité-Universitätsmedizin Berlin, Corporate Member of Freie Universität Berlin and Humboldt-Universität zu Berlin, Berlin, Germany;* [h] *Department of Urology, Erasmus MC, Rotterdam, The Netherlands;* [i] *Department of Urology, Institute of Clinical Sciences, Sahlgrenska Academy, University of Gothenburg, Gothenburg, Sweden;* [j] *Department of Urology, Region Västra Götaland, Sahlgrenska University Hospital, Gothenburg, Sweden;* [k] *Department of Urology, Netherlands Cancer Institute, Amsterdam, The Netherlands;* [l] *Department of Urology and Clinical Sciences, Danderyd Hospital, Danderyd, Sweden;* [m] *Department of Medical Epidemiology and Biostatistics, Karolinska Institute, Stockholm, Sweden;* [n] *Centre for Medical Imaging, University College London, London, UK;* [o] *Department of Radiology, Institute of Clinical Sciences, Sahlgrenska Academy, University of Gothenburg, Gothenburg, Sweden;* [p] *Department of Radiology, Sahlgrenska University Hospital, Gothenburg, Sweden;* [q] *Paul Strickland Scanner Centre, Mount Vernon Cancer Centre, Northwood, UK*

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Abstract

Background and objective: Prostate cancer (PCa) screening using prostate-specific antigen (PSA) thresholding and systematic biopsies reduces advanced disease presentations and cancer-specific mortality, but also leads to overdiagnosis. Magnetic resonance imaging (MRI) integration may maintain screening benefits, while reducing overdiagnosis and unnecessary biopsies. This review analyses the benefit-harm balance when MRI is integrated as first-line and second-stage (after PSA >3 ng/ml) test in PCa screening.
Methods: Following the Preferred Reporting Items for Systematic Reviews and Meta-analyses guidelines, we performed a PROSPERO-registered systematic review (CRD420251006926). Literature searches identified five first-line and four second-stage MRI screening studies. We assessed MRI strategies (first-line/second-stage and risk thresholds), biopsy avoidance, and biopsy methods (targeted/systematic) for histological outcomes (grade group [GG] ≥2/GG 1 cancer detection and benign biopsies). Benefit-to-harm ratios of >1 suggest a positive net benefit.
Key findings and limitations: First-line MRI screening detects twice as many men with GG ≥2 cancer as second-stage MRI screening but has more MRI-negative men (range, 66–89% vs 56–61%). Second-stage MRI significantly reduced biopsy rates (range, 42–

* Corresponding author. Department of Radiology and Nuclear Medicine, Erasmus University Medical Center Rotterdam, Postbus 2040, 3000 CA Rotterdam, The Netherlands. Tel. +31 6282 34355.
E-mail address: i.schoots@erasmusmc.nl (I.G. Schoots).

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79%) compared with systematic biopsy rates in all PSA-positive men. Subsequently, GG ≥2/GG 1 cancer detection ratios increased in MRI-positive men undergoing targeted and systematic biopsies (range, 1.9–6.2) and targeted biopsies alone (range, 1.8–7.0), compared with systematic biopsies alone (range, 0.8–1.4). First-line and second-stage MRI screening allowed biopsy avoidance in three to 55 and two to 15 men, respectively, for each benign diagnosis. All benefit-to-harm ratios showed positive net benefits (>1). Heterogeneity in the study protocols limits generalisability.

Conclusions and clinical implications: Targeted biopsies in second-stage MRI screening optimise clinically significant PCa detection, while reducing the number of biopsies. First-line MRI screening requires further assessments of its feasibility. PCa screening quality assurance requires standardised MRI interpretations and biopsy protocols.

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ADVANCING PRACTICE

What does this study add?

This systematic review analysed thoroughly the evidence regarding magnetic resonance imaging (MRI)-targeted biopsies in prostate cancer screening, emphasising the benefits and limitations of various MRI strategies. This review stresses the importance of standardised and robust quality control of biopsy protocols to enhance the selective detection of clinically significant prostate cancer while safely minimising unnecessary biopsies. It also underscores the need for clear thresholds for prostate-specific antigen and MRI to effectively guide screening pathways.

Clinical Relevance

The optimal strategy for prostate cancer screening continues to be the subject of considerable research. This comprehensive systematic review underscores the potential benefits of integrating prostate MRI into the diagnostic algorithm. Nonetheless, the role of MRI as a primary screening modality preceding PSA testing warrants further investigation. Two key conclusions can be drawn: first, rigorous quality assurance is critical throughout the diagnostic process, including both image acquisition/interpretation and MRI-informed biopsy procedures; second, risk-adapted diagnostic pathways should be prioritized to optimize the benefit-to-harm balance. Associate Edior: Gianluca Giannarini, MD.

Patient Summary

This review found that magnetic resonance imaging (MRI) scans can enhance the selective detection of significant prostate cancer cases while safely decreasing the number of biopsies in men undergoing prostate cancer screening. The effective use of MRI for screening necessitates careful planning and high standards for scanning and biopsy.

1. Introduction

Prostate cancer (PCa) remains a significant global health concern, ranking among the most frequently diagnosed cancers and a leading cause of cancer-related mortality in men [1]. This underscores the critical need for effective screening strategies to reduce mortality and improve patient outcomes [2].

Traditional PCa screening, which relies on prostate-specific antigen (PSA) testing followed by a systematic biopsy, can reduce the incidence of metastatic disease and cancer-specific mortality. However, this approach has several drawbacks. Untargeted systematic biopsies are associated with high rates of underdiagnosis of aggressive cancers, overdiagnosis of indolent cancers, unnecessary biopsy procedures, and overtreatments. These limitations have compromised the overall effectiveness and safety of traditional screening approaches, contributing to near-universal recommendations against population-wide PCa screening [3,4].

A multistep screening strategy incorporating clinical and biomarker-based risk stratification, magnetic resonance imaging (MRI) triage, and targeted biopsies may help optimise the balance between benefits and harms [5,6]. MRI with indicated targeted biopsies offers several advantages: it reduces unnecessary and unproductive procedures, limits the overdiagnosis of indolent cancers, improves the detection of aggressive disease, and enhances tumour grading [7]. A meta-analysis demonstrated a substantial reduction in biopsies through MRI-based screening compared with systematic biopsy pathways (odds ratio [OR]: 0.28 [95% confidence interval {CI}: 0.22–0.36]) [8]. Furthermore, by directing biopsies to MRI-identified suspicious areas (with no biopsy in MRI-negative men), the likelihood of detecting grade group (GG) 1 cancers was reduced by almost threefold (OR: 0.34 [95% CI: 0.23–0.49]).

This systematic review provides a comprehensive analysis of MRI-targeted biopsy approaches and their associated histological outcomes in the context of first-line and second-stage MRI-based PCa screening.

2. Methods

This systematic review has been registered in PROSPERO International Prospective Register of Systematic Reviews (CRD420251006926) and reported according to the 2020 Preferred Reporting Items for Systematic Reviews and Meta-analyses guidelines [9]. A comprehensive literature search was conducted using the MEDLINE, EMBASE, Web of Science, and Cochrane Central databases (Supplementary material). The target population were men within a PCa screening programme (population based or opportunistic) who underwent first-line MRI (without any PSA threshold) or second-stage MRI after a PSA threshold of 3 ng/ml. The target condition was GG ≥2 cancer detection.

Two independent reviewers (I.G.S. and A.R.P.) screened the titles, abstracts, and full texts for eligibility, and extracted the data. Data from duplicate reports were also searched for dataset completion. Discrepancies were resolved through discussions. The risk of bias was assessed using the Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) tool (Supplementary Fig. 1) [10]. A qualitative analysis was undertaken, and a statistical analysis was deemed not applicable.

The search identified studies evaluating the use of first-line [11–15] and second-stage [16–19] MRI. Studies were excluded if these focused on men with elevated risks based on germline mutations or race [20–24], repeated screening rounds [25–30], PSA thresholds of <3 ng/ml [31], and biomarkers other than PSA in addition to MRI (Supplementary Fig. 2) [32–37].

MRI strategies (first-line/second-stage and risk thresholds), biopsy avoidance, and biopsy methods (targeted/systematic) for histological outcomes (GG ≥2/GG 1 cancer detection and benign biopsies) were assessed. In addition, associations were explored between age, PSA positivity (thresholding at 3 ng/ml), and cancer detection rate (thresholding at GG ≥2). A descriptive analysis was carried out because the heterogeneous data were not appropriate for a statistical meta-analysis. Benefit-to-harm ratios with different descriptors and interpretations were calculated to compare the MRI-based screening and biopsy approaches (Table 1). Benefit-to-harm ratios of >1 suggest a positive net benefit.

3. Results

The two distinct strategies of first-line and second-stage MRI screening showed variability in participant recruitment and age inclusion (Table 2 and Fig. 1A and 1B), MRI acquisition and interpretation (Supplementary Table 1), and biopsy testing (Table 3), resulting in variable outcomes (Table 4 and Fig. 1C and 1D). The benefit-to-harm ratios (Table 5 and Fig. 2) were consistently above 1, showing the enhanced efficiency and reduced biopsy burden associated with the MRI-based screening approach.

3.1. *Prostate cancer screening approaches*

3.1.1. *First-line MRI screening*

First-line MRI, implemented as a primary screening tool without PSA testing, has the potential to detect clinically significant prostate cancer (csPCa) missed by serum PSA thresholding [12]. Approximately half of all GG ≥2 cancer cases were detected in men with PSA levels <3 ng/ml [12–14]. Most of the detected cancers are small GG 2 cancers,

Table 1 – Benefit-to-harm ratios with descriptors and interpretations, placed in the context of first-round population-based prostate cancer screening

Measurement	Description	Focus
Ratios for men undergoing biopsy		
Ratio: GG ≥2 PCa/GG 1 PCa	Shows the balance between detection of clinically significant or aggressive cancers (GG ≥2 PCa) and indolent cancers (GG 1 PCa)	Biopsy selectivity for clinically significant cancers
	A higher ratio indicates a better balance, detecting more aggressive cancers while minimising the detection of less aggressive cancers	
Ratio: GG ≥2 PCa/GG 1 PCa + benign biopsy	Shows the balance between detection of clinically significant or aggressive cancers (GG ≥2 PCa) and biopsy harm (ie, indolent cancers [GG 1 PCa] and benign biopsies)	Biopsy efficiency
	A higher ratio suggests fewer benign biopsies and lower detection rates of GG 1 PCa, indicating a more efficient biopsy approach	
Ratios for men avoiding biopsy		
Ratio: avoided biopsy/ undetected GG ≥2 PCa	Shows the balance between avoided biopsies (in men with negative MRI results) and undetected clinically significant or aggressive cancers (GG ≥2 PCa)	Safe avoidance
	A higher ratio suggests that the strategy successfully avoids biopsies without significantly increasing the risk of missing aggressive cancers	
Ratio: avoided biopsy/ benign biopsy	Shows the balance between avoided biopsies (in men with negative MRI results) and benign biopsies (in men with positive MRI results)	Selective avoidance of unproductive biopsies
	A higher ratio indicates a more favourable balance, as more men avoid unnecessary procedures while maintaining a good detection of aggressive cancers	
Important considerations in the ratios		
Context of use	Interpretation of these ratios depends on the clinical context (population-based screening or individual early diagnosis)	Varies by context
Role of MRI	Ratios vary based on whether MRI is used as the sole detection tool (first-line MRI) or as a triage tool after PSA testing (second-stage MRI)	Varies by approach
Definition of clinically significant cancer	The definition is crucial for interpreting the ratios, as the distinction between indolent and aggressive cancers impacts the assessment of benefits and harms. The definition is evolving to higher grading with improved diagnostic accuracy of MRI	Crucial to ratio interpretation

GG = grade group; MRI = magnetic resonance imaging; PSA = prostate-specific antigen.

Table 2 – Recruitment strategies and participation in first-round MRI prostate cancer screening studies

Study name	Author	Year	Recruitment process	Age inclusion (yr)	Recruitment period (yr)	Invited	Responded	Eligible (%)	Undergoing testing	PSA ≥3 ng/ml (%)	Age screened Median (IQR)
First-line MRI screening											
MVP [11]	Nam	2022	Men recruited from newspaper and radio advertisements from the Greater Toronto area	≥50	Not reported	NR	1188	525		NA	68 (±7.3)
IP1-PROSTAGRAM [13]	Eldred-Evans	2021	Men recruited through 7 primary care practices that were part of the UK National Institute for Health Research Clinical Research Network, London area, UK	50–69	10/2018–5/2019	2034	411	408 (20)		NA	57 (53–61)
VISIONING [14]	Wetteraurer	2024	Men recruited through general practitioners, neighbouring clinics, and urologists in private practice referring interested men from Basel, Switzerland	50 (45 high risk)	1/2019–5/2023	NR	NR	269		NA	58 (53–64)
RE-IMAGINE [12]	Moore	2023	Men recruited through searchable databases at partner general practitioners' surgeries (8): North London (5), South London (2) and Ilford (1), Essex	50–75	10/2019–12/2020	2096	457	334 (16)		NA	62 (±7)
PROSA (experimental) [15]	Messina	2024	Recruitment of male employees of Azienda Policlinico Umberto I, Rome, Italy	49–69	9/2020–9/2022	NR	NR	175		NA	59 (46–69)
Second-stage MRI screening											
Göteborg-2, 1st round (reference) [16]	Hugosson	2022	Men recruited from Göteborg, Sweden, or its 10 surrounding municipalities, randomly sampled	50–60	2015–2020	37 887	18 686	17 980 (47%)	5 994	405 (7)	56 (52–59)
Göteborg-2, 1st round (experimental) [16]									11 986	796 (7)	56 (52–59)
STHLM3-MRI, 1st round [17]	Eklund	2021	Men recruited from Stockholm County, Sweden, randomly selected by Statistics Sweden	50–74	2/2018–3/2020	49 118	NR	12 750 (26)	12 750	1 532 (12)	66 (61–71)
PROBASE, 1st round [18]	Boschheidgen	2024	Men recruited by random samples from local population registers in 4 study areas: Düsseldorf, Hannover, Heidelberg, and Munich, Germany	45	2/2014–12/2019	400 000	46 642	46 495 (12)	23 376	186 (1)	45 (44–47)
OPT [19]	Bratt	2024	Men recruited from region Skåne, Västra Götaland, and Stockholm, Sweden	50	2020–2022	68 060	NR	23 855 (35)	23 855	696 (3)	50 (50–50)
PROSA (reference) [15]	Messina	2024	Recruitment of male employees of Azienda Policlinico Umberto I, Rome, Italy	49–69	9/2020–9/2022	NR	NR	173	NR	61	59 (46–69)

IQR = interquartile range; MRI = magnetic resonance imaging; NA = not applicable; NR = not reported; PSA = prostate-specific antigen.

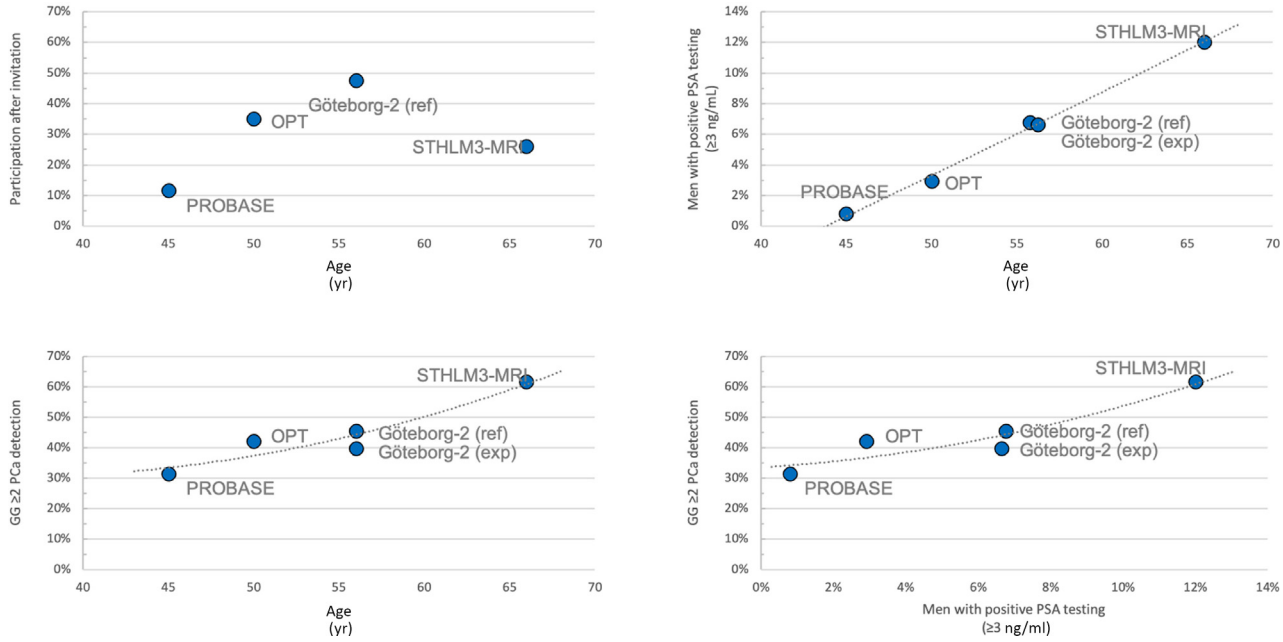


Fig. 1 – Second-stage MRI prostate cancer screening studies: (A) proportions of men participating in screening following invitation, categorised by age; (B) association between age and PSA positivity thresholding at 3 ng/ml; (C) link between age and cancer detection rate, with thresholding at grade group (GG) 2 and above; and (D) relationship between men with PSA positivity thresholding at 3 ng/ml and cancer detection rate, with thresholding at GG 2 and above. exp = experimental; MRI = magnetic resonance imaging; PCa = prostate cancer; PSA = prostate-specific antigen; ref = reference.

which are eligible for active surveillance and are unlikely to cause any harm if observed initially, thereby contributing to overdiagnosis [31]. Therefore, the clinical relevance of small-volume GG 2 cancers in men with PSA levels <3 ng/ml is debatable. GG ≥2 cancers are found rarely at PSA levels below 1 ng/ml [12–14]. Arguments against first-line MRI screening include higher cost, limited availability, and inter-reader variability [7,38]. Performing MRI on a large scale without prior risk stratification can result in a high proportion of negative scans due to the low prevalence of GG ≥2 cancers [39]. However, advancements in deep learning imaging reconstructions and deep learning–assisted image interpretations may help mitigate some of the limitations of first-line MRI screening [38].

3.1.2. *Second-stage MRI screening*

Second-stage MRI performed after PSA testing enriches the screened population for GG ≥2 cancers relative to the chosen PSA cut-off value. The primary role of MRI is to triage men for prostate biopsies. Men with negative or low-suspicion MRI findings can avoid an immediate biopsy. This selective biopsy strategy targets men at a higher risk based on their serum PSA levels and MRI findings. A key concern of PSA thresholding is the potential for a delayed diagnosis of clinically significant cancers, necessitating repeated PSA testing as a safety net [25,26].

3.2. *Target population for MRI-based screening*

The risk profile of individuals considered for MRI-based PCa screening is complex and influenced by various factors. Age, race, and family history are important considerations, and PSA level remains a crucial factor. Genetic predisposition and access to health care for underserved populations further complicate risk stratification [40]. This systematic review focuses on MRI-based strategies following PSA testing in broad age groups. A more universal approach to screening all eligible men, regardless of risk factors, may yield greater population benefits. The benefits and harms of the universal approach should be weighed against a personalised screening strategy based on multistep risk stratification, which has not yet been implemented [17,32].

3.2.1. *Age*

The age range in MRI screening trials typically spans from 45 to 75 yr (Table 2). The PROBASE trial focused on men aged 45 (median, 45; interquartile range [IQR], 44–47) yr [17], whereas the STHLM3-MRI study included men aged 50–74 (median, 66; IQR, 61–71) yr [15]. VISIONING included men aged 50 yr and higher-risk men aged 45 yr with African descent or a family history of PCa [14].

3.2.2. *High-risk men*

Men at a high risk of PCa include those of African descent and those with a strong family history. Black men exhibit a higher prevalence of GG ≥2 cancer, which often presents at a more advanced stage, emphasising the importance of early detection and risk stratification [41]. However, only one MRI-based screening study specifically focused on race has been reported (Table 2) [12].

3.2.3. *Family history and genetic predisposition*

Men with a strong family history or genetic predisposition also benefit from early screening [21,22,24]. Familial mutations in genes such as *BRCA2*, *ATM*, and *NBN* are associated with an increased risk of developing more aggressive PCa [42].

Table 3 – Overview of biopsy-based characteristics in first-round MRI-based prostate cancer screening studies

Study name	Biopsy sites, *n*	Biopsy strategy	Biopsy route	Biopsy method	Biopsy equipment/software	Biopsy protocol	Biopsy indication	Operators	Clinically significant cancer	Pathologists
First-line MRI screening										
MVP	1	Targeted and systematic	Transrectal	Image fusion	Artemis, Eigen	±4 targeted cores per target + 12-core systematic biopsy	MRI positive	NR	GG ≥2	Genitourinary cancer pathologists
IP1-PROSTAGRAM	1	Targeted and systematic	Transperineal	Image fusion	BiopSee, MedCom GmbH	Targeted cores	MRI positive	1	GG ≥2	Multiple expert uropathologists
VISIONING	1	Targeted and systematic	Transrectal/transperineal (general anaesthesia)	Image fusion	3D Guidance Trakstar/ Monalisa Biobot Surgical	2–3 targeted cores + 18-core systematic biopsy (small variations over 3 regions)	MRI positive and MRI negative with PSA >10 ng/ml or PSA density >0.15 ng/ml^2	NR	GG ≥2	NR
RE-IMAGINE	2	Targeted	Transrectal	Image fusion	NR	Targeted cores + systematic biopsy of peripheral zone	MRI positive and MRI negative with PSA density >0.12 ng/ml^2	Multiple	GG ≥2	NR
PROSA	1	Targeted	Transrectal	Image fusion	Urostation, Koelis	Targeted cores	MRI positive	NR	GG ≥2	1 genitourinary pathologist
Second-stage MRI screening										
Göteborg-2, 1st round	1	Targeted and systematic	Transrectal	Cognitive	BK Medical	10–12-core systematic biopsy ± 3–4 targeted cores per target	MRI positive	5 experienced urologists	GG ≥2	1 genitourinary pathologist, 2nd review by 2 external pathologists (blinded)
STHLM3-MRI, 1st round	4	Targeted and systematic	Transrectal	Image fusion	BK Medical	10–12-core systematic biopsy ± 2 targeted cores per target	MRI positive	Experienced urologists	GG ≥2	Centralised, 2–3 uropathologists
PROBASE, 1st round	4	Targeted and systematic	Transrectal and transperineal	Image fusion	Urostation Koelis; BiopSee Medcom; UroNAV Invivo	12–18-core systematic biopsy + targeted cores per target	MRI positive	NR	GG ≥2	1 genitourinary pathologist
OPT	3 regions	Targeted and systematic	NR	NR	NR	3–4 targeted cores per target, + 10–12-core systematic biopsy from dorsal sectors (small variations over 3 regions)	MRI negative: PI-RADS 1–2 + PSA density >0.15 ng/ml^2; MRI positive: PI-RADS 3 + PSA density >0.15 ng/ml^2,	NR	GG ≥2	16 sites, 3 regions

GG = grade group; MRI = magnetic resonance imaging; NR = not reported; PSA = prostate-specific antigen; PI-RADS = Prostate Imaging Reporting and Data System.

Table 4 – Outcomes of MRI-positive screens in MRI prostate cancer screening studies

Study name	Undergoing screen testing			Biopsy not indicated	Biopsy indicated		Biopsy results				Benefit-to-harm ratios [38]		
	Undergoing testing	PSA ≥3 ng/ml	MRI score 3–5	MRI-avoided biopsy (MRI score 1–2)	Biopsy not performed	Biopsy performed	Benign biopsy [a]	Any PCa detected	GG 1 PCa	GG ≥2 PCa	GG 2 PCa / GG 1 PCa	GG 2 PCa/GG 1 PCa + benign biopsy	Avoided biopsy/ benign biopsy
	*n*1	*n*2 (%) (/*n*1)	*n* (%) (/n2)	*n* (%) (/n2)	*n*3 (%) (/*n*2)	*n*4 (%) (/*n*2)	*n* (%) (/*n*3 + *n*4)	*n* (%) (/*n*3 + *n*4)	*n* (%) (/*n*2)	*n* (%) (/*n*2)			
First-line MRI screening													
Targeted + systematic biopsy in MRI-positive men													
MVP [b]	248	NA	25	221	1	24	4	15	4	11	**2.8**	**1.4**	**55**
			10	**90**	**4**	**96**	**17**	**63**	**17**	**46**			
IP1-PROSTAGRAM	408	NA	72	336	7	65	44	21	7	14	**2.0**	**0.3**	**8**
			18	**82**	**10**	**90**	**68**	**32**	**11**	**22**			
VISIONING	229	NA	77	152	0	77	48	29	8	21	**2.6**	**0.4**	**3**
			34	**66**	**0**	**100**	**62**	**38**	**10**	**27**			
Targeted biopsy in MRI-positive men													
RE-IMAGINE [c]	334	NA	48	286	17	31	4	27	2	25	**12.5**	**4.2**	**72**
			14	**86**	**35**	**65**	**13**	**87**	**6**	**81**			
IP1-PROSTAGRAM	408	NA	72	336	7	65	48	17	4	13	**3.3**	**0.3**	**7**
			18	**82**	**10**	**90**	**74**	**26**	**6**	**20**			
PROSA (experimental)	175	NA	20	155	10	10	4	6	2	4	**2.0**	**0.7**	**39**
			11	**89**	**50**	**50**	**40**	**60**	**20**	**40**			
Second-stage MRI screening													
Targeted + systematic biopsy in MRI-positive men													
Göteborg-2, 1st round (reference)	5994	405	142	263	12	130	40	90	31	59	**1.9**	**0.8**	**7**
		7	**35**	**65**	**8**	**92**	**31**	**69**	**24**	**45**			
STHLM3-MRI, 1st round	12 750	1 532	325	1 207	28	297	79	218	35	183	**5.2**	**1.6**	**15**
		12	**21**	**79**	**9**	**91**	**27**	**73**	**12**	**62**			
PROBASE, 1st round	23 376	186	107	79	18	89	48	41	13	28	**2.2**	**0.5**	**2**
		1	**58**	**42**	**17**	**83**	**54**	**46**	**15**	**31**			
OPT	23 855	696	236	460	15	221	84	137	44	93	**2.1**	**0.7**	**5**
		3	**34**	**66**	**6**	**94**	**38**	**62**	**20**	**42**			
VISIONING	229	51	28	23	0	30	15	15	2	13	**6.5**	**0.8**	**2**
		22	**55**	**45**	**0**	**107**	**50**	**50**	**7**	**43**			
Targeted biopsy in MRI-positive men													
Göteborg-2, 1st round (experimental)	11 986	796	267	529	6	261	98	163	59	104	**1.8**	**0.7**	**5**
		7	**34**	**66**	**2**	**98**	**38**	**62**	**23**	**40**			
STHLM3-MRI, 1st round	12 750	1 532	325	1 207	28	293	108	185	23	162	**7.0**	**1.2**	**11**
		12	**21**	**79**	**9**	**90**	**37**	**63**	**8**	**55**			
PROBASE, 1st round	23 376	186	107	79	18	89	59	30	9	21	**2.3**	**0.3**	**1**
		1	**58**	**42**	**17**	**83**	**66**	**34**	**10**	**24**			
IP1-PROSTAGRAM	408	41	10	31	1	9	3	6	0	6	NA	**2.0**	**10**
		10	**24**	**76**	**10**	**90**	**33**	**67**	**0**	**67**			
PROSA (reference)	173	61	9	52	3	6	4	2	2	0	NA	NA	**13**
		35	**15**	**85**	**33**	**67**	**67**	**33**	**33**	**0**			

GG = grade group; MRI = magnetic resonance imaging; NA = not applicable; PCa = prostate cancer; PI-RADS = Prostate Imaging Reporting and Data System; PSA = prostate-specific antigen.

[a] Benign biopsy included any noncancerous histology, including ASAP and HGPIN patterns, excluding all GG 1 and GG ≥2 cancers.

[b] The threshold for a positive MRI result was PI-RADS 4 and 5, instead of PI-RADS 3–5 as in other studies.

[c] The threshold for a positive MRI result was suspicious signal intensity on a high b value, instead of PI-RADS 3–5 as in other studies.

Table 5 – Comparison between first-line and second-stage MRI screening

Feature	First-line MRI screening	Second-stage MRI screening
Role of MRI	Primary screening tool, detecting prostate cancer regardless of PSA level	Utilised after PSA testing, where MRI is used to triage men for prostate biopsies
Target population	All men within a defined age range	Men with a PSA level above a predetermined threshold (eg, 3 ng/ml). Determining the optimal PSA threshold for MRI referral to balance cancer detection sensitivity with cost effectiveness is challenging (key concern)
Cancer detection rate	Lower overall cancer detection due to the absence of preselection. Detects cancer cases missed by PSA testing. GG ≥2 cancer detection rates range from 3% to 8%	Higher overall cancer detection due to preselection of men with elevated PSA. GG ≥2 cancer detection rates range from 14% to 21%
Biopsy rate	Potentially higher biopsy rate, due to more indeterminate MRI interpretations, owing to the low cancer prevalence in the (unselected and relatively young) population (key concern)	Overall, a lower biopsy rate, due to biopsy avoidance in men below the PSA threshold. By only targeting men with positive MRI results, biopsies in men with MRI negative results are further avoided
Cost effectiveness	Lower cost effectiveness is likely due to high MRI costs and the potentially high proportion of negative MRI results in populations with a low cancer prevalence (key concern)	Potentially higher cost effectiveness due to selective MRI use for men more likely to have aggressive (GG ≥2) cancers
Risk of delayed diagnosis	Potentially lower delayed diagnoses, as MRI detects aggressive cancers that may not elevate PSA	Present but delayed diagnosis is mitigated by repeat screening rounds (key concern)
Overdiagnosis	Likely higher overdiagnosis as more men will undergo MRI and biopsy testing. However, MRI is less sensitive for detecting indolent (GG 1) prostate cancer	Likely lower overdiagnosis as only the preselected men will undergo MRI and biopsy testing. Substantially reduced overdiagnosis compared with PSA-based screening with systematic biopsies
Deployment recommendations	MRI-only screening remains experimental and may require refinement	Favoured for immediate adoption due to established pathways, and cost efficiency, but to a lower benefit-to-harm ratio

GG = grade group; MRI = magnetic resonance imaging; PSA = prostate-specific antigen.

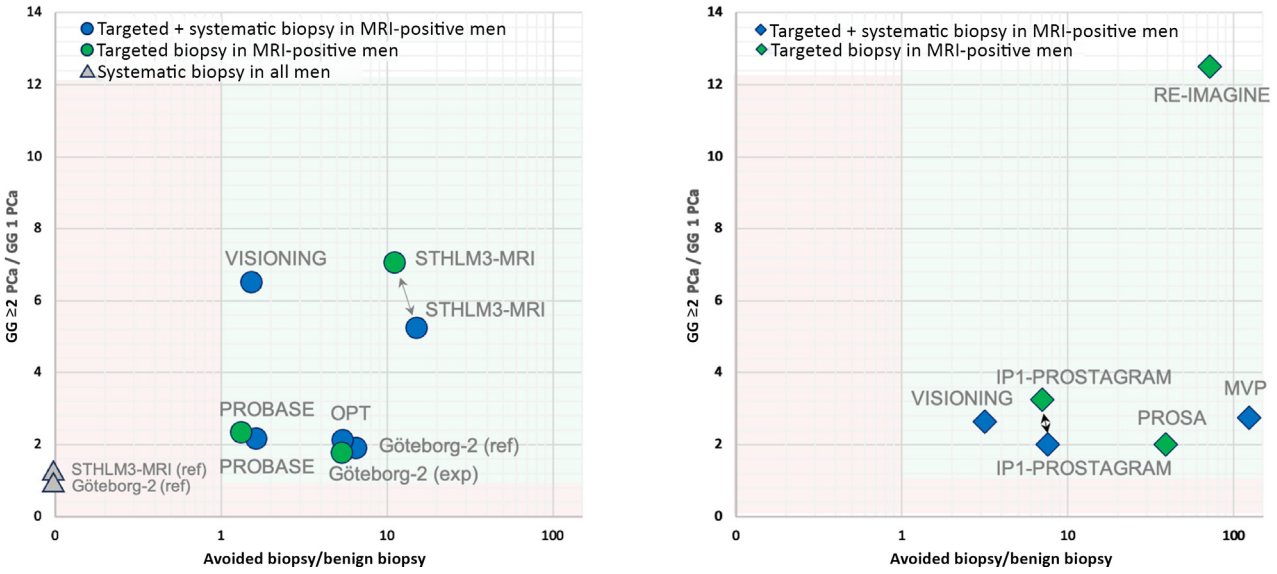


Fig. 2 – Benefit-to-harm ratio assessments in MRI-based prostate cancer screening studies. The ratio of grade group (GG) ≥2 cancer to GG 1 cancer detection (benefit-to-harm ratio) is plotted on the Y axis, indicating biopsy selectivity. The ratio of avoided biopsies to benign (unproductive) biopsies (benefit-to-harm ratio) is plotted on the X axis, indicating the selective avoidance of biopsies. (A) Plotted ratios from Table 4 of studies in which MRI was performed after PSA testing (second-stage testing). The "targeted and systematic biopsy" strategy (blue circles) and the "targeted biopsy" strategy (green circles) in MRI-positive men show a positive net benefit (>1; light green background) in GG ≥2/GG 1 cancer detection ratio, indicating higher biopsy selectivity for clinically significant cancers and more selective biopsy avoidance of unnecessary biopsies. The traditional "systematic biopsy in all men" strategy (grey triangles; ie, Göteborg-2 and STHLM3-MRI) shows net harm due to less selective biopsies (light red background). (B) Plotted ratios from Table 4 of first-line testing studies. Both the "targeted and systematic biopsy" strategy (blue diamonds) and the "targeted biopsy" strategy (green diamonds) in MRI-positive men show a positive net benefit for cancer detection selectivity (higher GG ≥2/GG 1 cancer detection ratio) and more selective avoidance of unnecessary biopsies (avoided/benign biopsy ratio). exp = experimental; MRI = magnetic resonance imaging; PCa = prostate cancer; PSA = prostate-specific antigen; ref = reference.

3.2.4. *Participation rates*

Screening participation is influenced by factors such as age, information provision, investigative burden, risk stratification, trial accessibility, geographical and demographic characteristics, population diversity, and prior screening experience. The highest participation rate was observed in Göteborg-2 (47%; median, 56 yr; IQR, 53–59; Table 2). OPT reported a 35% participation rate for the period 2020–2022. The participation rates in countries with diverse races remain undocumented. The association between participation and age suggests a lower interest in younger men (Fig. 1A). Adherence to biopsy recommendations following

abnormal MRI findings was high. The no-biopsy rates in MRI-positive screened men were 6%, 8%, and 9% in the OPT, Göteborg-2, and STHLM3-MRI studies, respectively.

3.3. Screen test characteristics

When MRI is used as a second-stage test, its function is to reduce the number of unnecessary biopsies.

3.3.1. PSA thresholds for MRI referral

Data from first-line MRI screening studies indicate that a PSA cut-off of 3 ng/ml would miss half of all GG ≥2 cancer cases in the first round of screening [12–14]. This finding is corroborated by Göteborg-2 (second-stage MRI), where men with baseline PSA levels between 1.8 and 3.0 ng/ml had GG ≥2 cancers in 4.6% (31/670), mostly GG 2 cancers [31]. MRI in men with PSA levels between 1.8 and 3.0 ng/ml did not increase the detection of GG 1 cancers. However, the use of a PSA threshold of 1.8 ng/ml instead of 3 ng/ml would result in a 2.6-fold increase in MRI utilisation [31].

The second-round, 2-yr interval data from Göteborg-2 and STHLM3-MRI demonstrated that the majority of GG ≥2 cancer cases occurred in men with PSA levels <3.0 ng/ml in the first round (approximately 68% in both trials) [25,26,43]. The diagnostic delay until the PSA level reached 3 ng/ml did not appear to compromise cure rates, as metastatic disease was not detected, and only two men had (microscopic) T3 disease on prostatectomy pathology [31]. The oncological safety of rescreening until the PSA level reaches 3 ng/ml requires further investigation [43].

Age-specific PSA thresholds could help ensure that MRI is used in men who are more likely to benefit [2]. The association between PSA positivity and age (Fig. 1B) suggests that age-specific PSA thresholds could further optimise MRI utilisation, especially in younger men with lower PSA levels who are less likely to have GG ≥2 cancer. The TRANSFORM trial will evaluate the optimal PSA threshold for MRI-based risk-adapted screening in different MRI arms (PSA none, and >1 and >3 ng/ml) [44].

PSA levels above the threshold of 3 ng/ml in young men (aged 45–50 yr) should be confirmed, since half of the men in PROBASE did not have a persistently elevated PSA level at repeat testing [45]. This proportion was 17% in STHLM3-MRI in men aged 50–69 yr [46].

3.3.2. PSA-density threshold for MRI referral

The PSA density cut-off values for MRI referral in screening have not been investigated widely in first-round screening trials. This is likely due to the added complexity of upfront prostate volume estimation via digital rectal examinations or ultrasound. OPT and Göteborg-2 evaluated PSA density cut-offs (using prostate volumes from previous MRI scans) to trigger MRI referrals in subsequent rounds, but specific data have not yet been published. Such data may also emerge from European screening studies supported by the PRAISE-U project [47].

3.3.3. MRI threshold for biopsy referral

There is an emerging consensus on the use of abbreviated MRI protocols in the screening context [38]. The Prostate Imaging Reporting and Data System (PI-RADS) assessment system is not ideal for abbreviated protocols or screening. PI-RADS was developed for multiplanar, multiparametric prostate MRI assessments in the diagnostic setting of a higher disease prevalence [48]. Provided that MRI scans are of high quality and experienced readers undertake interpretations, the following PI-RADS–based observations can be deduced for the screening context [38].

3.3.3.1. Negative MRI results. A PI-RADS category score of 1 or 2 indicates that csPCa is unlikely. Patient-level GG ≥2 cancer detection rates after a negative MRI result are very low in screening. In the reference arm of Göteborg-2, nine cases of GG ≥2 cancer were detected in 240 men undergoing systematic biopsies after negative MRI results [16]. These were all GG 2 cancer cases; six had <5% Gleason pattern 4, representing favourable intermediate-risk disease. In the repeat screening report of Göteborg-2, the GG ≥2 cancer detection rate in men who had negative MRI results at the initial screening was similar between those who previously underwent a systematic biopsy (reference arm: 3.2% [12/371]) and those who did not (experimental arm: 2.6% [11/421]) [25]. Similarly, in the repeat screening report of STHLM3-MRI, the detection rate of GG ≥2 cancer was only 3.9% in men with negative MRI results at initial screening; these were also predominantly GG 2 cancer cases [26]. Therefore, an MRI suspicion score of 1 or 2 should be categorised as a negative screening result [38].

For second-stage MRI, the frequency of scores of 1 or 2 was approximately 60% (eg, OPT, 59%; STHLM3-MRI, 56%; Göteborg-2 control arm, 59%; and Göteborg-2 experimental arm, 61%; Supplementary Table 1) [16,17,19]. The frequency of negative MRI results increases in first-line MRI screening (range, 66–89%) [11,13–15].

3.3.3.2. Negative MRI result with high PSA or PSA density. Some screening trials have incorporated systematic biopsies in men with negative MRI results due to elevated serum PSA levels. For instance, VISIONING employed biopsies in cases of abnormal digital rectal examination results and/or PSA levels above 10 ng/ml despite a negative MRI result [14]. Furthermore, a PSA density of >0.15 ng/ml^2 was a criterion for a systematic biopsy in cases with negative MRI results. The RE-IMAGINE (>0.12 ng/ml^2) and OPT (>0.15 ng/ml^2) studies also considered PSA density as a screen-positive result despite negative MRI findings [12,19]. However, cancer yields and unproductive biopsy rates in this elevated-risk category have not been reported.

3.3.3.3. Indeterminate and positive MRI results. An indeterminate MRI result (category score 3) was observed in approximately 20% of second-stage MRI screening interpretations (range, 8–43%; Supplementary Table 1), which matches the diagnostic setting [49,50]. This category is more frequent in younger men (PROBASE 43% and OPT 21%), who generally have a lower PCa prevalence [18,19]. Management of men with indeterminate results should generally favour biopsy avoidance with a return to screening, as protocolised in OPT [19]. If needed, multiplanar and contrast-enhanced sequences can serve as secondary assessments to avoid biopsies [12,51]. A high PSA density can also aid in biopsy decisions for men with indeterminate

MRI results [52,53]. In OPT, men with a PI-RADS score of 3 and PSA density above 0.15 ng/ml^2 underwent MRI-targeted biopsies [19]. In the "interactive report" [54] from OPT, the cancer detection rates in such men were 25% for GG 1 (26/106), 32% for GG 2–3 (34/106) and 3.8% for GG 4–5 (four of 106) cancer. However, a retrospective analysis of both screening rounds of STHLM3-MRI found that PSA density was not effective in risk stratifying PI-RADS 3 cases [55]. Therefore, recommending a PSA density–based biopsy strategy should be approached with caution.

A suspicion score of 4 or 5 can be categorised as an MRI-positive screen, indicating the need for a biopsy. The frequency of PI-RADS score 4 (15%; range, 7–28%) in screening was greater than that of PI-RADS score 5 (ranged, 4–8%), indicating that MRI-detected lesions are generally smaller (Supplementary Table 1).

3.4. Post–screen test strategies

Two distinct biopsy strategies have emerged for men with positive MRI results: targeted biopsies alone and targeted biopsies with additional systematic biopsies (Table 3).

3.4.1. MRI-targeted biopsy approaches

Variability in the biopsy route (transrectal or transperineal), biopsy method (image fusion or cognitive), and biopsy equipment (stepper approach, tracking field, or freehand) was observed across the screening trials (Table 3). In most studies, the number of biopsy operators was not limited, and the experience of biopsy-performing operators was not specified. Furthermore, the number of systematic biopsies varied from 10 to 18. Targeted biopsy protocols were more controlled, allowing two to four cores per target, primarily at the discretion of the biopsy operator. Biopsy protocols were often not specified further regarding target lesion size, prostate volume, or location (apex vs base, urethra, or anterior-posterior).

3.4.2. Additional systematic to targeted biopsies in MRI-positive men

A direct comparison within Göteborg-2, comparing targeted and systematic biopsies with targeted biopsies alone, showed reductions in GG $\geq$2 cancer detection rates from 45% to 40% in MRI-positive men (Table 4) [16]. Similarly, the PROBASE and STHLM3-MRI studies showed decreases in GG $\geq$2 cancer detection rates from 31% to 24% and from 62% to 55%, respectively [17,18]. These data suggest that a systematic biopsy may add value to the screening context, especially in young men with lower-volume prostate glands, indeterminate MRI readings, and smaller-sized category 4 lesions [56]. However, the meta-analysis of screening trials did not find a significant benefit of a systematic biopsy for detecting additional cases of GG $\geq$2 (OR: 1.07, p = 0.07) or GG $\geq$3 (OR: 1.06, p = 0.43) cancers [8].

Göteborg-2, which compared targeted and systematic biopsies with targeted biopsies alone, showed no decrease in GG 1 cancer detection rate (from 24% to 23%; Table 4). However, reduced GG 1 cancer detection was more pronounced in the PROBASE and STHLM3-MRI studies, with GG 1 cancer detection rate decreasing from 15% to 10% and from 12% to 8%, respectively. These data suggest that systematic biopsies may contribute to overdiagnosis. A meta-analysis also confirmed higher GG 1 cancer detection (OR: 1.16, p = 0.01) with systematic sampling [8]. Based on the GG $\geq$2 and GG 1 cancer detection results, the addition of systematic biopsies to targeted biopsies in MRI-positive men in screening should be considered carefully during biopsy planning, particularly for younger men and those with smaller lesions.

Given the observed variability in biopsy approaches and the observation that systematic biopsies can detect additional cases of GG $\geq$2 cancer, standardised prebiopsy planning is crucial. This should encompass organ and target segmentations, explicitly defining the number and placement of biopsy cores, including perilesional sampling [57]. The safest biopsy route, tailored to the lesion location and patient factors, must be selected within this framework.

4. Discussion

4.1. Target outcomes: benefits and harms

The term csPCa distinguishes cancers that can cause patient morbidity and mortality from those that rarely do. This distinction is critical because "clinically insignificant" PCa is common and overtreatment can lead to potentially harmful side effects. Clinically significant PCa is commonly defined using simple histological subtypes [58]. Studies utilising MRI in early diagnosis and screening settings generally report the detection rates of GG 1, GG 2, and GG $\geq$3 cancers, using these as surrogates of clinical significance [59,60].

4.1.1. GG $\geq$2 cancer detection (benefit)

In second-stage MRI screening, the GG $\geq$2 cancer detection rates in men undergoing targeted and systematic biopsies were generally higher than in first-line MRI screening (range, 31–62% vs 22–27%), likely due to the higher disease prevalence by PSA preselections (Table 4). This trend in GG $\geq$2 cancer detection is less clear in men undergoing targeted biopsies only (range, 24–67% vs 20–81%).

The wide variation in GG $\geq$2 cancer detection rates across studies cannot be explained solely by differences in MRI utilisation and MRI-targeted biopsy approaches. Age differences in study inclusion (Fig. 1C) and differences in PSA positivity (Fig. 1D) also contributed by affecting the GG $\geq$2 cancer prevalence. Other impactful features include variability in MRI interpretation, particularly in younger men [18], and variations in biopsy procedures and histopathological interpretations.

4.1.2. GG 1 cancer detection (harm)

The detection rate of GG 1 cancer was generally higher in second-stage MRI screening than in first-line MRI screening (range, 7–24% vs 10–11%; Table 4), probably related to the higher biopsy rates.

4.1.3. Benign biopsies (harm)

Benign biopsy rates in second-stage MRI screening using an MRI-targeted and systematic approach were lower than those in first-line MRI screening (range, 27–54% vs. 62–68%; excluding *MVP* data, which only biopsied men with PI-RADS 4-5 results [11]; Table 4).

4.1.4. Biopsy avoidance (benefit)

Avoided biopsies are a key benefit of MRI use in PCa screening. In second-stage MRI screening, biopsy avoidance rates due to MRI were lower than in first-line MRI screening (range, 42–85% vs 66–89%; Table 4). This is likely related to the lower number of MRI-tested men and a subsequent higher disease prevalence in the second-stage approach, due to the PSA thresholding effect. However, when focusing on the total screened population, biopsy avoidance rates were higher in the second-stage approach (88–99.5% vs 66–89%).

4.2. Benefit-to-harm ratios as summary outcome measures

Standardised metrics are essential for comparing different screening methods and evaluating the impact of MRI. Table 2 summarises the key differences between MRI first-line and second-stage approaches, and Table 3 outlines the measurement ratios of biopsy precision and avoidance for methodology comparisons [38,61]. The benefit-to-harm ratios must be contextualised to the screening goal (individual diagnosis vs population impact), screening round, MRI integration in the pathway, chosen MRI cut-offs for screen-positivity [12], and the definition of clinically significant cancer explains these descriptors in the context of first-round population-based screening.

Our analysis of Göteborg-2 and STHLM3-MRI demonstrated consistent improvements in the benefit-to-harm ratios of MRI-based screening (Table 4). Specifically, the ratio of GG $\geq$2 to GG 1 cancer cases among biopsied men improved substantially, indicating increased selectivity for detecting GG $\geq$2 cancer cases (Fig. 2). The benefit-to-harm balance remained positive when unproductive biopsies were also considered harmful. Furthermore, the significantly higher ratio of biopsies avoided after negative MRI to unproductive biopsies after positive MRI underscores the enhanced efficiency and reduced biopsy burden associated with the MRI-based screening approach.

4.2.1. Ratio of GG $\geq$2 to GG 1 cancer detection (benefit-to-harm measure)

For second-stage MRI screening, GG $\geq$2/GG 1 cancer detection ratios in men undergoing targeted and systematic biopsies range from 1.9 to 6.2 (Table 4), demonstrating a net positive benefit (Fig. 2A). The ratio is lower for first-line MRI screening (Fig. 2B), ranging from 2.0 to 2.8, likely due to the reduced prevalence of disease.

A similar trend was observed in men undergoing targeted biopsies only, with ratios ranging from 1.8 to 7.0 for second-stage MRI (Table 4 and Fig. 2), indicating a limited advantage for additional systematic biopsies. In a direct comparison between the two arms in Göteborg-2, contrasting targeted and systematic biopsies with targeted biopsies alone, ratios of 1.9 and 1.8, respectively, were observed, indicating minimal additional benefits from systematic biopsies. In contrast, the STHLM3-MRI study showed an increased ratio from 5.2 to 7.0, which was attributable to a lower incidence of GG 1 cancer cases detected solely by targeted biopsies.

4.2.2. Ratio of avoided biopsies to unproductive biopsies (benefit-to-harm measure)

For second-stage MRI screening, the ratio of avoided to unproductive biopsies in men undergoing targeted and systematic biopsies ranged from 2 to 15, providing substantial evidence for a net positive benefit-to-harm ratio regarding the selective avoidance of biopsies (Table 4 and Fig. 2). These ratios were higher in first-line MRI screening, ranging from 3 to 55, due to the relatively higher number of negative MRI results.

4.3. Risk of bias

While the risk of bias was generally low in the second-stage MRI screening studies, some risk of bias was identified in the first-line MRI screening studies, predominantly related to patient selections (Supplementary Fig. 1). Additionally, a publication bias appeared to be limited.

4.4. Limitations

Several limitations must be considered when interpreting our findings. Firstly, the relatively small number of studies that met the inclusion criteria, coupled with their focus predominantly on Western European populations, limits the broader generalisability of our findings to other ethnicities and geographic regions. The included studies exhibited heterogeneity in the age range of participants, with typical enrolment spanning 30 yr from 45 to 75 yr. The changing appearance of the prostate gland and the increasing prevalence of cancer will affect pathological outcomes. Secondly, heterogeneity in the study protocols across the included studies further compounds this limitation. Most studies did not perform systematic biopsies as a comparative reference standard in men with negative MRI results. The negative predictive value of MRI is not 100% and varies across settings. This omission precluded the assessment of population-level diagnostic performance metrics, such as sensitivity and specificity, which is why we emphasised the biopsy outcomes in men with positive MRI results. Furthermore, owing to this heterogeneity, pooled metrics of diagnostic performances would not be methodologically appropriate. Third, we excluded studies focusing on populations at a higher risk of PCa, such as those with germline mutations. The lack of specific data about these populations restricts the broad applicability of the review's conclusions. Furthermore, our approach does not address the potential of personalised screening strategies that incorporate multistep risk stratification based on a combination of clinical factors, diverse biomarkers, and individual life expectancy. This further limits the scope of this review in the context of evolving multimodal risk assessment paradigms.

Several factors related to MRI acquisition and interpretation also warrant consideration as limitations. These are covered in detail in a companion review [38]. In summary, substantial inter-reader variability in the interpretation of MRI findings has been documented, which could introduce heterogeneity across the included studies. Although PI-RADS is used widely, its optimal application and interpretation in a screening context remain a subject of on-going

refinement. Moreover, variations in MRI image quality and the specific scoring systems used across different studies were not assessed systematically, potentially contributing to the variability in reported pathological outcomes.

Finally, the long-term oncological outcomes of MRI-based PCa screening strategies remain unclear. Although preliminary data suggest the potential for extending screening intervals based on initial PSA levels and MRI findings, further investigation is required to establish the long-term safety and efficacy of such approaches. Moreover, the clinical significance of screen-detected, small-volume GG 2 cancers is unknown, also highlighting the lack of a universal consensus on a vital pathological definition to assess the impact of MRI on the outcomes of screening.

5. Conclusions

This review comprehensively analysed the integration of MRI and MRI-targeted biopsies into PCa screening programmes. Implementation of MRI and targeted biopsies in screening requires attention to well-defined screening positivity thresholds based on MRI and PSA levels, to reduce variability and ensure consistent screening outcomes. Optimisation and quality control of MRI-targeted biopsy techniques should be prioritised to minimise the undersampling of suspected, primarily small, MRI-detected lesions. The addition of systematic biopsies to targeted biopsies in MRI-positive men should be considered carefully during biopsy planning, particularly for younger men and those with smaller lesions, and higher PSA/PSA-density values. Ultimately, successful integration of MRI and targeted biopsies in screening will require radiologists and urologists to refine protocols, standardise biopsy procedures, and embrace advanced MRI technologies. Ideally, diagnostic workups should be conducted in dedicated centres with multidisciplinary expertise. Such initiatives will help maximise benefits while minimising harms and facilitate greater acceptance of PCa screening.

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Study concept and design: Schoots, Padhani.
Acquisition of data: Schoots, Padhani.
Analysis and interpretation of data: Schoots, Padhani.
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Supplementary material

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